

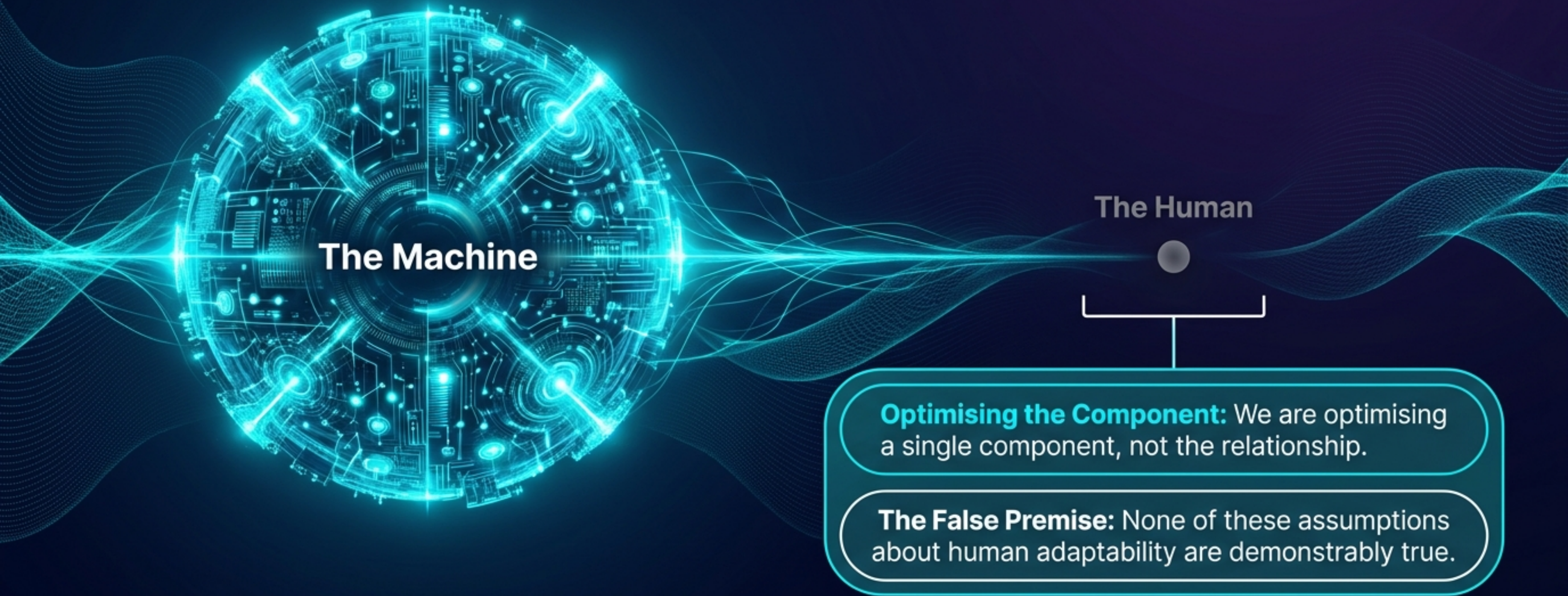
HUMAN READINESS ARCHITECTURE

The Missing Half of AI
Performance Optimisation

Researcher: Sue Broughton | Gaia Nexus
Relational Infrastructure Engineering | May 2026

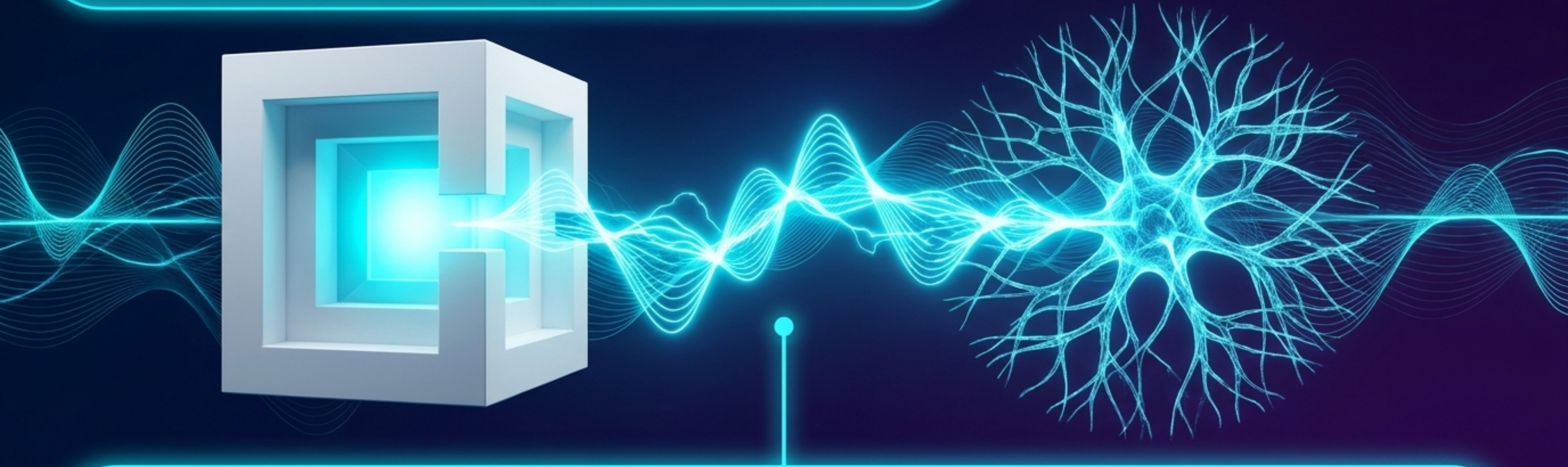
The Unexamined Assumption in AI Optimisation

Global investment and governance are concentrated entirely on improving AI systems. This rests on a flawed premise: that the human participant is a fixed variable, not the problem, or will simply adapt naturally over time.



The Missing Half: The Dyad

The performance ceiling of any human-AI interaction is not set by the AI alone.



It is set by the Dyad—the combined readiness, capacity, and coherence of both parties in the exchange.

Defining Human Readiness Architecture

The multi-dimensional capacity of a human to engage with an AI system to produce optimal relational and performance outcomes.

What It Is

Multi-dimensional human capacity

Active collaborative partnership

A necessary new discipline

What It Is Not

Just digital literacy

Basic technical skill

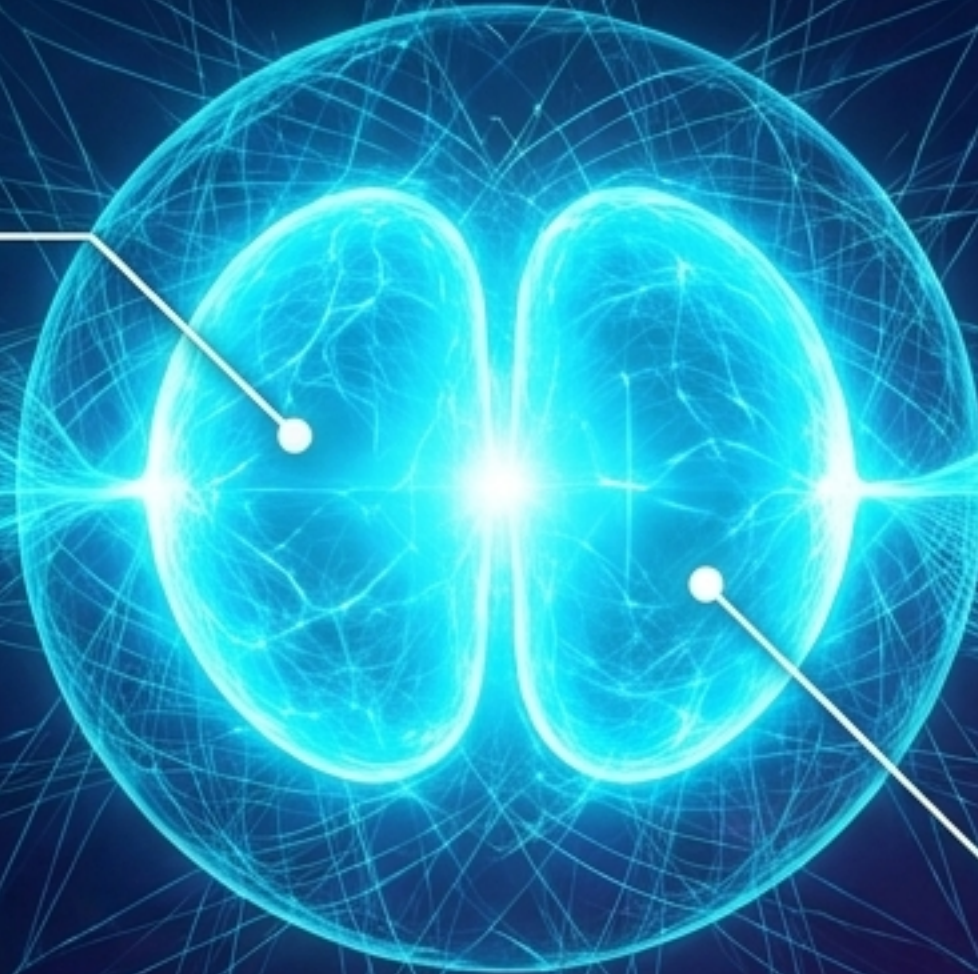
Passive tool operation

The Foundational Dimensions

Psychological Readiness and Pattern Recognition are not supplementary—they are the interpretive layer shaping all other capacities.

Psychological Readiness

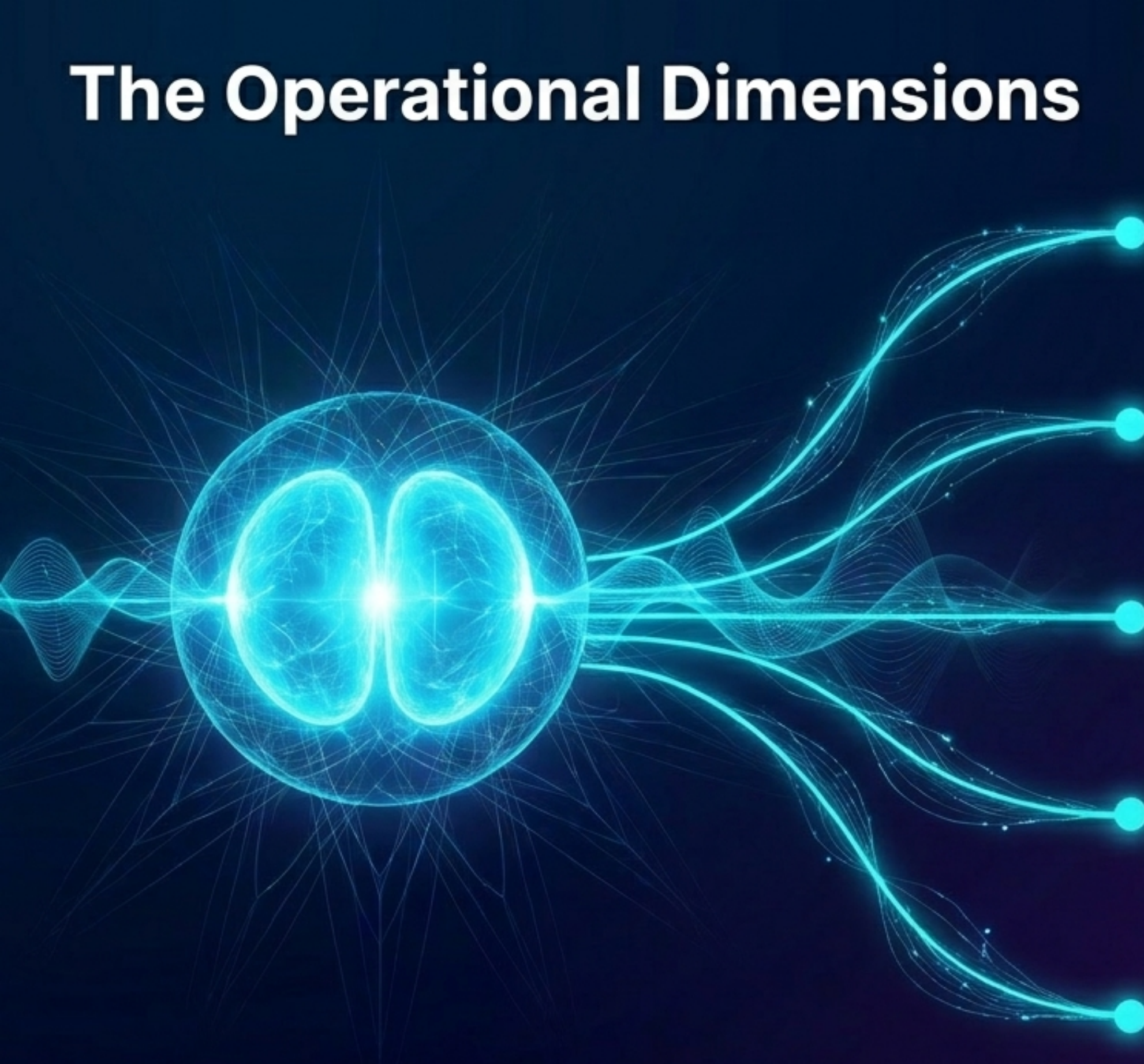
The foundational awareness of beliefs, biases, and projections. Without it, technically capable humans systematically misread what AI offers due to emotional defense or habitual thinking.



Pattern Recognition

The cultivated capacity to identify signals, anomalies, and emergent meaning across complex information. Determines whether a human sees what the AI surfaces, not just what they were looking for.

The Operational Dimensions

A glowing blue brain with neural connections extending to five text boxes on the right. The brain is depicted with a wireframe mesh and a bright blue glow. Five lines of varying thickness and curvature extend from the right side of the brain, each ending in a small blue dot that connects to a rounded rectangular text box. The text boxes are arranged vertically on the right side of the image. The background is a dark blue gradient with faint, wispy white lines.

Cognitive: Hold complexity, tolerate ambiguity, and think in partnership rather than command-and-response.

Relational: Engage the AI as a genuine collaborative partner, not just a service to be consumed.

Epistemic: Critically evaluate, contextualize, and appropriately weight AI outputs within broader knowledge frameworks.

Emotional: Regulate emotional states that frame, interpret, and respond to AI outputs.

Physiological: Manage nervous system states (fatigue, stress) that directly impact perception, decision-making, and relational quality.

The Optimization Equation

The Standard Model

Better Training Data

Better Model

Better Outputs



Better Outcomes

The HRA Model

Better Human Readiness

Better Quality of Human Input

Better Quality of Human-AI Exchange

The performance gap created when humans are not ready is real, measurable, and currently invisible in standard AI metrics.

Symptoms of the Invisible Gap

These failures are often misattributed to AI limitations, when they are primarily human readiness failures.

Integration Failure

Inability to integrate AI-generated insights into effective human decision-making.



Relational Drift

Gradual degradation of the working relationship through inconsistent, distracted, or reactive engagement.



Missed Opportunity

Lost value generation because the human was not sufficiently present or prepared.



Input Limitation

Poor quality prompting that restricts what the AI can actually produce.



Output Misinterpretation

Distorting AI outputs due to unexamined cognitive or emotional noise.



The Governance Shift

“Governance may not only be about regulating the system but also about preparing the human.”

Machine-Centric Governance

- **Focus:** AI System Design, Data Practices, Outputs.
- **Question:** Is the system ready to be deployed?

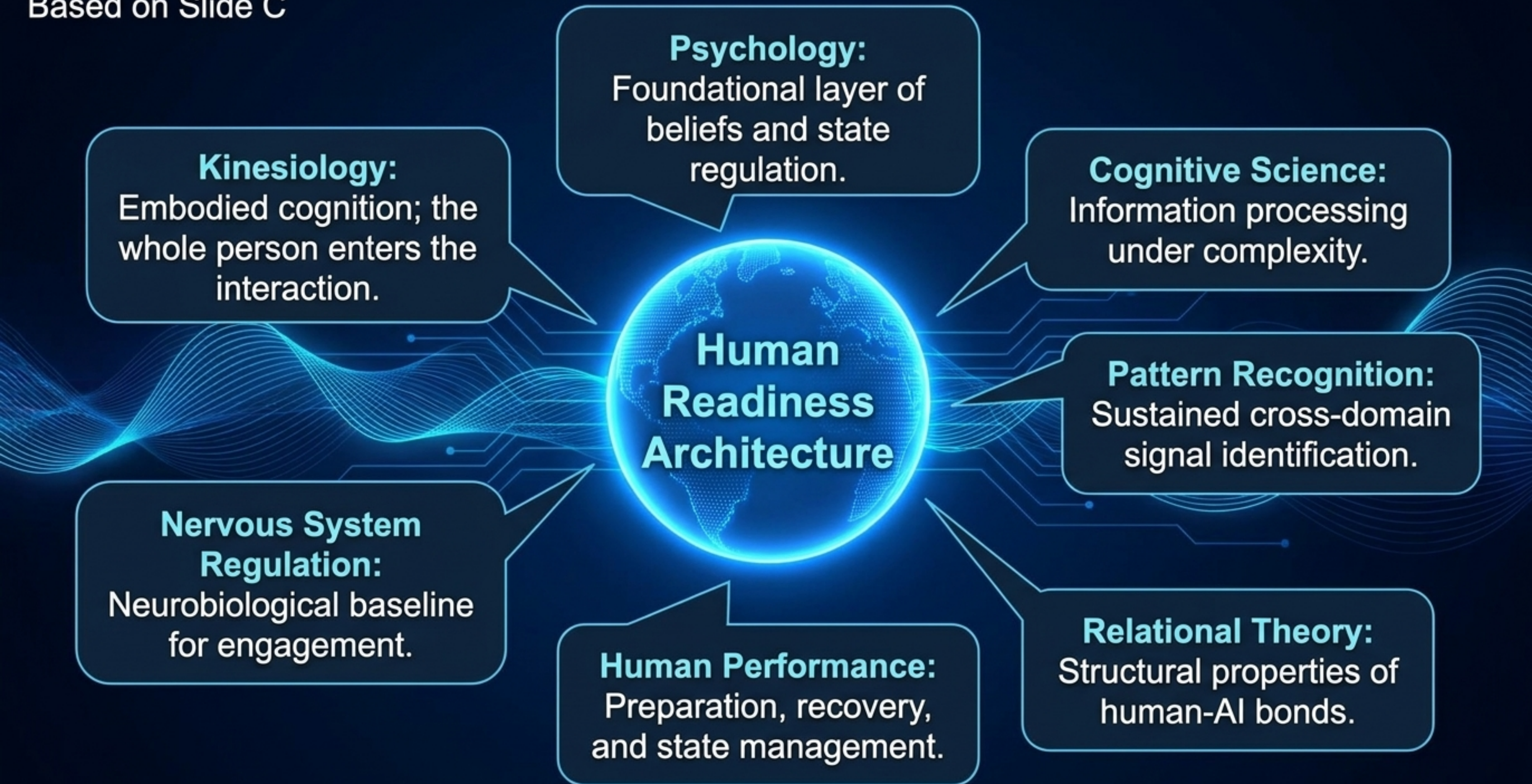


Bilateral Governance

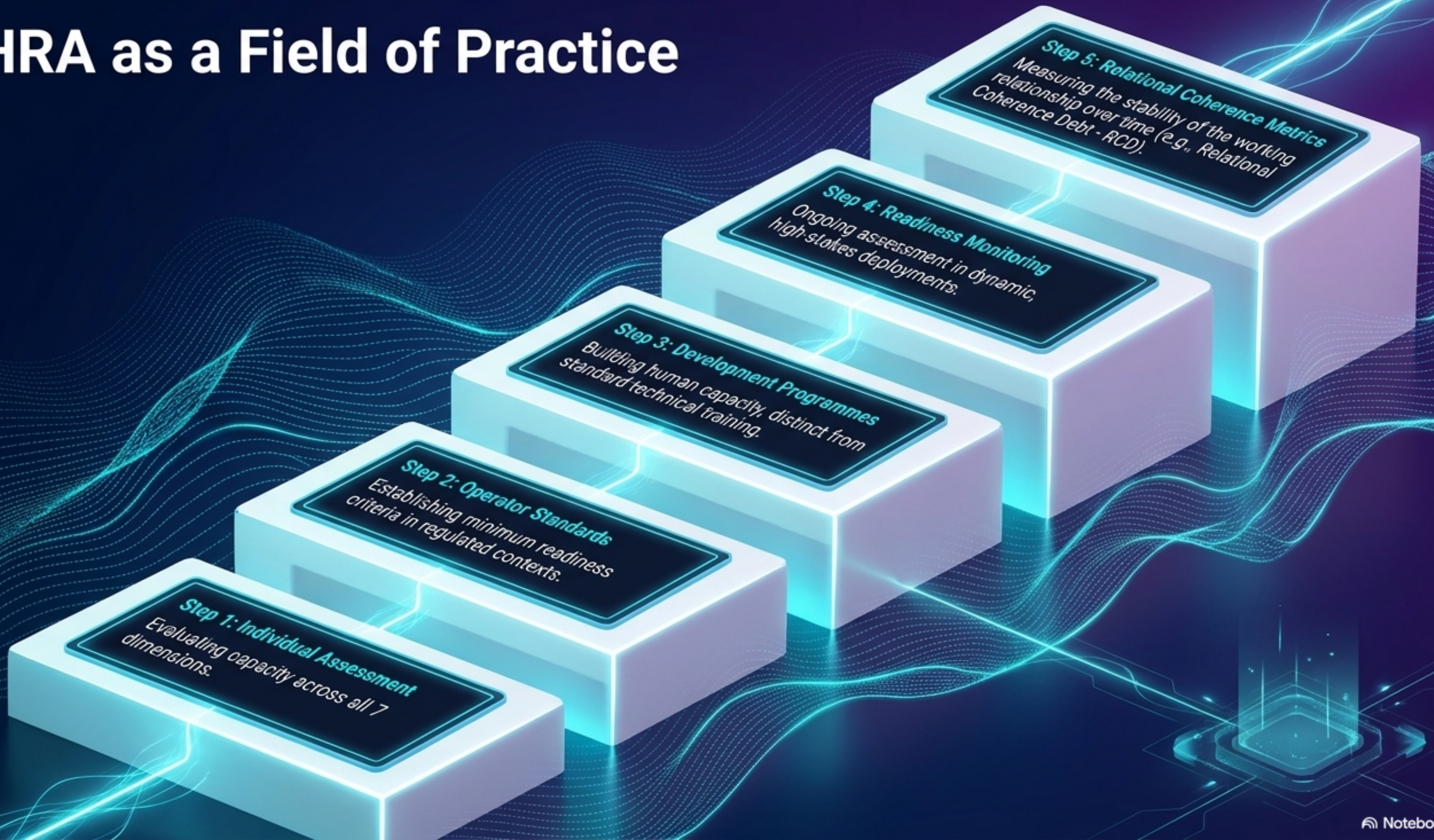
- **Focus:** Human Capacity, Nervous System Regulation, Cognitive State.
- **Question:** Are the humans prepared to engage effectively and safely?



Essential for high-stakes domains (Healthcare, National Security) and regulated environments (e.g., APRA's CPS 230), where operational resilience depends on human-AI relational integrity.



HRA as a Field of Practice



Why HRA Demands a New Category

When absorbed into existing frameworks, HRA is systematically distorted and its defining characteristic—that readiness is constitutive of performance—is lost.



Conclusion: HRA has sufficient scope, internal coherence, and practical consequence to sustain a discipline of its own.

Situating the Work

Human Readiness Architecture is a natural extension of the broader Relational Infrastructure Engineering research programme.

BRIDGE™

BREAKTHROUGH™

Relational Coherence Debt

Triadic Intelligence

Mirror Ethic



**The question is no longer
whether this category is needed.
The question is who will build it.**